Yanyan Dong

yanyandong@arizona.edu

RESEARCH INTEREST

Natural language processing, human-AI interaction, and intelligent system design, with a focus on making AI systems more accessible, interpretable, and useful for real-world users.

EDUCATION

University of Arizona *Aug 2025 - Present*
Ph.D. in Information, Advisor: Dr. Steven Bethard

University of Arizona *2025*
Master of Science, Information

PROJECTS

Agent Visualization for Human-AI Collaboration, University of Arizona *Sept 2025 - Present*
Advisors: Dr. Sandra Bae, Dr. Steven Bethard

- **Objective:** Design human-AI collaborative systems that support user oversight and informed decision-making in agentic workflows.
- Developing human-centered interfaces that support user decision-making when collaborating with AI agents.
- Conducting user studies to investigate how interface design impacts user comprehension, trust, and decision quality when working with AI agents.

Research Opportunity Success System, University of Arizona *Jan 2024 - Present*
Research Assistant, Advisors: Misael Cabrera

- **Objective:** Develop a generative AI-based system to identify highly relevant mining and minerals research opportunities for faculty and researchers.
- Designed an end-to-end system architecture.
- Created and administered surveys to identify faculty needs and refine system design based on feedback.
- Built a data pipeline to collect, organize, and store funding opportunity data in a structured database for retrieval.
- Developed a retrieval-augmented generation (RAG) system that leverages state-of-the-art dense retrieval models and large language models (LLMs) to summarize retrieved grant opportunities and report relevance based on researcher interests.
- Designed and developed a web application prototype integrating data collection, RAG, and results presentation to enhance usability.
- **Tech:** Python, Hugging Face Transformers, Scikit-Learn, PyTorch, SQL, Sentence Transformers

AI-Powered Safety System for Mining Industry, University of Arizona *Jan 2023 - Present*
Advisors: Dr. Leonard D. Brown

- **Objective:** Develop an AI-powered system that provides safety recommendations tailored to specific mining work environments.
- Processed millions of tokens from raw text datasets and conducted domain-specific pretraining of transformer encoder-based models (e.g., RoBERTa). Fine-tuned these models on a 16k dataset to build classifiers for detecting potential worker injury types.
- Developed a prompting framework leveraging LLMs (e.g., GPT-4, Claude 3 Opus) for in-context learning to identify possible injury types and generate corresponding safety recommendations.
- Employed few-shot prompting and RAG to incorporate domain-specific mining safety knowledge, improving LLM performance on specialized tasks.

- Conducted comparative analyses of model results, visualized findings, and authored papers summarizing key insights.
- Developing an LLM-based verification pipeline for validating generated safety recommendations.
- **Tech:** Python, Hugging Face Transformers, Scikit-Learn, PyTorch, OpenAI APIs, Anthropic APIs

Temporal Information Extraction, University of Arizona *Jan 2024 - May 2024*
Advisor: Dr. Steven Bethard

- **Objective:** Extract event-event, event-time, and time-time temporal relations from unstructured text.
- Preprocessed standard benchmark datasets (e.g., MATRES, TimeBank Dense) into various formats for different task formats, such as pairwise sequence classification and natural language inference.
- Fine-tuned RoBERTa models for multiple temporal relation extraction tasks, establishing baselines.
- Implemented an efficient inference pipeline utilizing open-source LLMs (e.g., Flan-T5, Vicuna-7b, Mistral-7b) and performed prompt engineering using VLLM library.
- Fine-tuned multiple large models (e.g., Flan-T5, Vicuna-7b, Mistral-7b) using Low-Rank Adaptation (LoRA) for optimized performance.
- **Tech:** Python, Hugging Face Transformers/PEFT, Scikit-Learn, PyTorch, VLLM

Topic Analysis of National Science Foundation (NSF) Grant Proposals, University of Arizona *Aug 2023 - Dec 2023*
Advisor: Dr. Bryan Heidorn

- **Objective:** Investigate the evolution of NSF grant research topics over the past two decades.
- Conducted exploratory data analysis (e.g., document length, noise analysis) and performed data preprocessing.
- Developed topic modeling approaches (e.g., LDA) on NSF grant proposal data and analyzed the resulting topics.
- Applied clustering algorithms to proposals represented via Bag-of-Words and pretrained transformer embeddings, then conducted manual annotation to assess cluster quality.
- Visualized topic modeling results using pyLDAvis and Word Clouds, and performed manual interpretation of identified topics.
- **Tech:** Python, Hugging Face Transformers, Scikit-Learn, PyTorch, pyLDAvis, WordCloud, Sentence Transformers

PUBLICATIONS & PRESENTATIONS

Yanyan Dong and Leonard D. Brown. Applying Generative AI to Job Hazard Analysis and Safety Observation Reports. *MINEXCHANGE SME Annual Conference & Expo 2025 (manuscript)*

Yanyan Dong, Leonard D. Brown, Hong Cui and Jeff Burgess. Injury Prediction and Root Cause Analysis Using Large Language Models. *MINEXCHANGE SME Annual Conference & Expo 2024 (manuscript)*

AWARDS

Distinguished Graduate Scholar, College of Information Science, University of Arizona *2025*

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash, HTML/CSS, LaTeX
Frameworks & Libraries: PyTorch, Hugging Face Transformers, Scikit-learn, Keras, Sentence Transformers, PEFT/LoRA, vLLM, React, Bootstrap, Firebase
Tools & Platforms: Git, Docker, Linux, Slurm, REST APIs, Tableau, Power BI, Figma

Languages: English, Mandarin, Cantonese